

PREACT 8/9 PRINTABLE PRACTICE EXAM 02 - MATH

35 Questions | 40 Minutes

Directions: Choose the best answer for each question. Mark only one answer per question. The answer key and explanations begin after the question section.

QUESTIONS

M1-001

Which of the following numbers is greatest?

- A. -0.82
- B. -0.78
- C. -0.75
- D. -0.70
- E. -0.71

M1-002

What value of x satisfies $5x - 7 = 18$?

- A. 2
- B. 5
- C. 7
- D. $11/5$
- E. 25

M1-003

Two angles form a linear pair. If one angle measures 68° , what is the measure of the other angle?

- A. 22°
- B. 68°
- C. 102°
- D. 112°
- E. 248°

M1-004

A tank contains 50 gallons of water at time $t = 0$. Water drains at a constant rate. The table shows the amount W of water in the tank at several times.

t (minutes)	0	2	4	6
W (gallons)	50	42	34	26

Which equation gives W in terms of t ?

- A. $W = 50 - 4t$
- B. $W = 50 - 8t$
- C. $W = 50 + 4t$
- D. $W = 4t$
- E. $W = 25 - 4t$

M1-005

What is the median of the data set 4, 6, 7, 9, 14?

- A. 6
- B. 7
- C. 8
- D. 8.5
- E. 9

M1-006

What is $(3.2 \times 10^5)(4 \times 10^{-2})$ written in scientific notation?

- A. 1.28×10^3
- B. 1.28×10^4
- C. 1.28×10^7
- D. 12.8×10^4
- E. 12.8×10^7

M1-007

A jacket has an original price of \$80. It is discounted by 25%, and then 5% sales tax is applied to the discounted price. What is the final price?

- A. \$59
- B. \$60
- C. \$63
- D. \$64
- E. \$84

M1-008

A recipe uses 2.5 cups of flour to make 20 muffins. At the same rate, how many cups of flour are needed to make 34 muffins?

- A. 3.4 cups
- B. 4 cups
- C. 4.25 cups
- D. 5 cups
- E. 6.8 cups

M1-009

A rectangular playground is 6 meters wide and 8 meters long. A straight path is built from one corner to the opposite corner. How long is the path?

- A. 7 meters
- B. 10 meters
- C. 12 meters
- D. 14 meters
- E. 48 meters

M1-010

A fair spinner is divided into 8 equal sections numbered 1 through 8. What is the probability that the spinner lands on an even number?

- A. $\frac{1}{8}$
- B. $\frac{1}{4}$
- C. $\frac{3}{8}$
- D. $\frac{1}{2}$
- E. $\frac{5}{8}$

M1-011

What is the value of $\sqrt{81} + \sqrt[3]{64}$?

- A. 7
- B. 11
- C. 12
- D. 13
- E. 17

M1-012

The first term of an arithmetic sequence is 7, and each term after the first is 4 greater than the preceding term. What is the eighth term?

- A. 32
- B. 35
- C. 39
- D. 56
- E. 63

M1-013

A club has \$260 to spend on a trip. A reservation fee costs \$44, and each student ticket costs \$18. What is the greatest number of student tickets the club can buy without exceeding its budget?

- A. 10
- B. 11
- C. 12
- D. 13
- E. 14

M1-014

Point P has coordinates $(3, -2)$. Point P is rotated 90° counterclockwise about the origin. What are the coordinates of the image of P?

- A. $(-3, 2)$
- B. $(-2, -3)$
- C. $(-2, 3)$
- D. $(2, -3)$
- E. $(2, 3)$

M1-015

The table shows whether 120 students participate in band.

	In band	Not in band	Total
Grade 8	18	42	60
Grade 9	27	33	60
Total	45	75	120

One student is chosen at random from the students who are in band. What is the probability that the chosen student is in Grade 9?

- A. $\frac{3}{20}$
- B. $\frac{9}{40}$
- C. $\frac{2}{5}$
- D. $\frac{3}{5}$
- E. $\frac{3}{4}$

M1-016

A printer prints 18 pages in 45 seconds at a constant rate. How many pages does it print in 5 minutes?

- A. 90
- B. 100
- C. 108
- D. 120
- E. 135

M1-017

At a school play, 23 adult and student tickets were sold for a total of \$188. Adult tickets cost \$11 each, and student tickets cost \$6 each. How many student tickets were sold?

- A. 8
- B. 10
- C. 12
- D. 13
- E. 15

M1-018

A laboratory culture begins with 40 cells. The number of cells triples every 2 hours. How many cells are in the culture after 6 hours?

- A. 120
- B. 360
- C. 720
- D. 1,080
- E. 3,240

M1-019

A cylindrical grain container has an interior radius of 3 meters and an interior height of 5 meters. What is its interior volume?

- A. 15π cubic meters
- B. 30π cubic meters
- C. 45π cubic meters
- D. 75π cubic meters
- E. 90π cubic meters

M1-020

A school wants to estimate the percentage of all its students who are satisfied with the cafeteria. Which sampling method is most likely to produce a representative sample?

- A. Survey the first 100 students who enter the cafeteria on Monday.
- B. Survey every student on the basketball teams.
- C. Randomly select 25 students from each grade.
- D. Ask students to complete an optional survey posted on the cafeteria door.
- E. Survey 100 students who bring lunch from home.

M1-021

What is the value of $(2^3 \times 2^5) \div 2^4$?

- A. 2
- B. 4
- C. 8
- D. 16
- E. 256

M1-022

The equation $x^2 - 7x + 12 = 0$ has two solutions. What is the greater solution?

- A. 2
- B. 3
- C. 4
- D. 6
- E. 12

M1-023

A taxi ride costs \$3.50 plus \$2.25 for each mile traveled. If a rider can spend at most \$17.00, what is the greatest whole number of miles the rider can travel?

- A. 4
- B. 5
- C. 6
- D. 7
- E. 8

M1-024

Two triangles are similar. The side lengths of the smaller triangle are 6, 9, and 12 centimeters. The shortest side of the larger triangle is 10 centimeters. What is the length of the side of the larger triangle that corresponds to the 9-centimeter side?

- A. 12 centimeters
- B. 13.5 centimeters
- C. 15 centimeters
- D. 18 centimeters
- E. 20 centimeters

M1-025

A linear model $h = 1.8a + 52$ estimates a plant's height h , in centimeters, after a weeks. According to the model, what height is predicted after 12 weeks?

- A. 64.0 centimeters
- B. 70.2 centimeters
- C. 72.0 centimeters
- D. 73.6 centimeters
- E. 74.8 centimeters

M1-026

A rectangular patio is 12 meters long and 8 meters wide. A path 1 meter wide is built around the outside of all four sides of the patio. Paving the path costs \$18 per square meter. What is the total paving cost?

- A. \$396
- B. \$612
- C. \$720
- D. \$792
- E. \$2,520

M1-027

What value of x satisfies $3(2x - 5) - 4 = 2(x + 7) + 3$?

- A. 4
- B. 6
- C. 8
- D. 9
- E. 12

M1-028

Which of the following numbers is irrational?

- A. $\sqrt{49}$
- B. 0.121212...
- C. $5/11$
- D. $\sqrt[3]{27}$
- E. $\sqrt{18}$

M1-029

What is the distance between the points $(-2, 1)$ and $(4, 9)$ in the coordinate plane?

- A. 8
- B. 10
- C. 12
- D. 14
- E. 100

M1-030

A linear function p passes through the points $(-2, 7)$ and $(4, -5)$. What is $p(1)$?

- A. -1
- B. 0
- C. 1
- D. 3
- E. 5

M1-031

One integer is selected at random from the integers 1 through 30. What is the probability that the selected integer is a multiple of 4 or a multiple of 6?

- A. $1/5$
- B. $4/15$
- C. $1/3$
- D. $2/5$
- E. $7/15$

M1-032

An aquarium has a rectangular base that is 60 centimeters by 30 centimeters. Water is added at a constant rate, causing the water depth to rise from 12 centimeters to 20 centimeters in 6 minutes. At what rate is water added? Use 1,000 cubic centimeters = 1 liter.

- A. 1.8 liters per minute
- B. 2.0 liters per minute
- C. 2.4 liters per minute
- D. 8.0 liters per minute
- E. 14.4 liters per minute

M1-033

A school store buys 48 items consisting only of notebooks and binders. Each notebook costs \$2.50, each binder costs \$6.50, and the total cost is \$188. How many binders does the store buy?

- A. 14
- B. 16
- C. 17
- D. 24
- E. 31

M1-034

A scale model of a rectangular storage room uses a scale of 1:40. The model's interior dimensions are 12 centimeters by 7.5 centimeters by 5 centimeters. What is the actual interior volume of the storage room in cubic meters?

- A. 7.2 cubic meters
- B. 18 cubic meters
- C. 28.8 cubic meters
- D. 72 cubic meters
- E. 288 cubic meters

M1-035

A delivery company recorded these delivery times, in minutes, for 5 packages: 18, 20, 20, 22, 25. A sixth delivery time of 40 minutes is added to the data set. Which statement correctly compares the mean and median before and after the sixth time is added?

- A. The mean increases by 1 minute, and the median increases by $19/6$ minutes.
- B. The mean increases by $19/6$ minutes, and the median increases by 1 minute.
- C. The mean increases by 4 minutes, and the median does not change.
- D. The mean and median each increase by $19/6$ minutes.
- E. The mean and median each increase by 1 minute.

ANSWER KEY + EXPLANATIONS

Each explanation corresponds to the numbered question in the practice section.

M1-001 - Correct: D. -0.70

Explanation: Among negative numbers, the number closest to 0 is greatest. Of the choices, -0.70 is closest to 0.

M1-002 - Correct: B. 5

Explanation: Add 7 to both sides to get $5x = 25$, and then divide by 5 to get $x = 5$.

M1-003 - Correct: D. 112°

Explanation: Angles in a linear pair sum to 180° , so the other angle measures $180^\circ - 68^\circ = 112^\circ$.

M1-004 - Correct: A. $W = 50 - 4t$

Explanation: The tank starts with 50 gallons. Its amount decreases by 8 gallons every 2 minutes, so the rate is -4 gallons per minute and $W = 50 - 4t$.

M1-005 - Correct: B. 7

Explanation: The data are already ordered, and the middle of the 5 values is 7.

M1-006 - Correct: B. 1.28×10^4

Explanation: Multiply the coefficients and powers: $3.2 \times 4 = 12.8$ and $10^5 \times 10^{-2} = 10^3$. Thus, $12.8 \times 10^3 = 1.28 \times 10^4$.

M1-007 - Correct: C. \$63

Explanation: The discounted price is $\$80 \times 0.75 = \60 . Applying 5% tax gives $\$60 \times 1.05 = \63 .

M1-008 - Correct: C. 4.25 cups

Explanation: The unit rate is $2.5 \div 20 = 0.125$ cup per muffin. For 34 muffins, $34 \times 0.125 = 4.25$ cups.

M1-009 - Correct: B. 10 meters

Explanation: The path is the hypotenuse of a right triangle with legs 6 and 8. Its length is $\sqrt{6^2 + 8^2} = \sqrt{100} = 10$ meters.

M1-010 - Correct: D. $1/2$

Explanation: Four of the 8 equally likely numbers, 2, 4, 6, and 8, are even, so the probability is $4/8 = 1/2$.

M1-011 - Correct: D. 13

Explanation: $\sqrt{81} = 9$ and $\sqrt[3]{64} = 4$, so $\sqrt{81} + \sqrt[3]{64} = 9 + 4 = 13$.

M1-012 - Correct: B. 35

Explanation: Seven increases of 4 occur from the first term to the eighth term, so the eighth term is $7 + 7 \times 4 = 35$.

M1-013 - Correct: C. 12

Explanation: After the \$44 fee, \$216 remains. Since $\$216 \div \$18 = 12$, the club can buy at most 12 student tickets.

M1-014 - Correct: E. (2, 3)

Explanation: A 90° counterclockwise rotation maps (x, y) to $(-y, x)$. Thus, $(3, -2)$ maps to $(2, 3)$.

M1-015 - Correct: D. $3/5$

Explanation: Of the 45 students in band, 27 are in Grade 9. The probability is $27/45 = 3/5$.

M1-016 - Correct: D. 120

Explanation: Five minutes is 300 seconds. At 18 pages per 45 seconds, the printer produces $(18/45) \times 300 = 120$ pages.

M1-017 - Correct: D. 13

Explanation: Let a and s be the numbers of adult and student tickets. Then $a + s = 23$ and $11a + 6s = 188$. Substituting $a = 23 - s$ gives $253 - 5s = 188$, so $s = 13$.

M1-018 - Correct: D. 1,080

Explanation: Six hours contains 3 two-hour periods. Tripling 3 times gives $40 \times 3^3 = 40 \times 27 = 1,080$ cells.

M1-019 - Correct: C. 45π cubic meters

Explanation: A cylinder's volume is $\pi r^2 h$. With $r = 3$ and $h = 5$, the volume is $\pi \times 3^2 \times 5 = 45\pi$ cubic meters.

M1-020 - Correct: C. Randomly select 25 students from each grade.

Explanation: Randomly selecting students from every grade includes the whole school population while reducing grade-level selection bias.

M1-021 - Correct: D. 16

Explanation: Using exponent rules gives $2^{3+5-4} = 2^4 = 16$.

M1-022 - Correct: C. 4

Explanation: Factoring gives $x^2 - 7x + 12 = (x - 3)(x - 4)$, so the solutions are 3 and 4. The greater solution is 4.

M1-023 - Correct: C. 6

Explanation: The cost must satisfy $3.50 + 2.25m \leq 17.00$. Thus, $2.25m \leq 13.50$, so $m \leq 6$; the greatest whole number is 6.

M1-024 - Correct: C. 15 centimeters

Explanation: The scale factor from the smaller triangle to the larger triangle is $10/6 = 5/3$. The corresponding side is $9 \times 5/3 = 15$ centimeters.

M1-025 - Correct: D. 73.6 centimeters

Explanation: Substitute $a = 12$ into the model: $h = 1.8(12) + 52 = 21.6 + 52 = 73.6$ centimeters.

M1-026 - Correct: D. \$792

Explanation: The outer dimensions are 14 meters by 10 meters, so the path's area is $(14 \times 10) - (12 \times 8) = 140 - 96 = 44$ square meters. The cost is $44 \times \$18 = \792 .

M1-027 - Correct: D. 9

Explanation: Expanding gives $6x - 19 = 2x + 17$. Then $4x = 36$, so $x = 9$.

M1-028 - Correct: E. $\sqrt{18}$

Explanation: $\sqrt{49} = 7$, $0.121212\ldots$ is a repeating decimal, $5/11$ is a ratio of integers, and $\sqrt[3]{27} = 3$, so those four numbers are rational. Since 18 is not a perfect square, $\sqrt{18}$ is irrational.

M1-029 - Correct: B. 10

Explanation: The horizontal change is 6 and the vertical change is 8. The distance is $\sqrt{6^2 + 8^2} = \sqrt{100} = 10$.

M1-030 - Correct: C. 1

Explanation: The slope is $(-5 - 7)/(4 - (-2)) = -12/6 = -2$. Using $(-2, 7)$ gives $p(x) = -2x + 3$, so $p(1) = 1$.

M1-031 - Correct: C. $1/3$

Explanation: There are 7 multiples of 4 and 5 multiples of 6 from 1 through 30. The 2 common multiples, 12 and 24, were counted twice, so there are $7 + 5 - 2 = 10$ favorable integers. The probability is $10/30 = 1/3$.

M1-032 - Correct: C. 2.4 liters per minute

Explanation: The depth increases by 8 centimeters, so the added volume is $60 \times 30 \times 8 = 14,400$ cubic centimeters, or 14.4 liters. Over 6 minutes, the rate is $14.4 \div 6 = 2.4$ liters per minute.

M1-033 - Correct: C. 17

Explanation: If all 48 items were notebooks, they would cost \$120. Each binder replacing a notebook adds \$4, and $\$188 - \$120 = \$68$. Therefore, the number of binders is $\$68 \div \$4 = 17$.

M1-034 - Correct: C. 28.8 cubic meters

Explanation: Multiplying each model dimension by 40 gives actual dimensions of 480, 300, and 200 centimeters, or 4.8, 3, and 2 meters. The volume is $4.8 \times 3 \times 2 = 28.8$ cubic meters.

M1-035 - Correct: B. The mean increases by $19/6$ minutes, and the median increases by 1 minute.

Explanation: Originally, the mean is $105 \div 5 = 21$ and the median is 20. With 40 added, the mean is $145/6$, an increase of $145/6 - 21 = 19/6$, and the median is $(20 + 22) \div 2 = 21$, an increase of 1.